

AI USAGE DISCLOSURE FORM

ACIS - Agentic Code Intelligence

Samsung PRISM GenAI Hackathon · Theme 1

1. Team Details

Team Name	SRMIST_JARVIS
Project / Product Name	ACIS - Agentic Code Intelligence
Organization / Institution	SRM INSTITUTE OF SCIENCE AND TECHNOLOGY
Submission Date	04 October 2026

2. AI Usage Declaration

Did the team use Artificial Intelligence (AI) in developing this project? YES

AI tools were used as development assistance for architecture brainstorming, code generation/refactoring support, UI/UX iteration, documentation/content drafting, and testing/debugging. The team reviewed, modified, integrated, and validated the resulting work against the ACIS project requirements and measured system behavior.

3. Purpose of AI Usage

Purpose	How AI was used
Idea generation / brainstorming	No, we referred various official opensources for architecture exploration, retrieval strategy discussion, failure-mode analysis, and demo planning, but we used ai for brainstorming, but we didn't get the complete end-end idea from ai.
Code generation or assistance	Yes, implementation assistance, refactoring, debugging, and development of supporting components/scripts.
UI / UX design	No, used official documents for interface structure, presentation of retrieval evidence, evaluation views, version/evolution views, and iterative UI refinement.
Content creation	No, we came up with the ideation and some ai tech to refine our technical explanations, presentation/demo speech drafts, README/documentation assistance, and disclosure drafting.
Data analysis	Yes, for comparative analysis, but maximum we used official sources for assistance interpreting retrieval metrics, baseline comparisons, per-query results, and evaluation outputs.
Testing / debugging	Yes, for test planning, diagnosis of implementation issues, and interpretation of test/evaluation failures.
Other	Architecture review, implementation planning, prompt engineering, and technical review of the ACIS retrieval pipeline, we used AI for our guidance, as our mentor, entire implementation strategy is we discussed and finalized, we used ai for correctness of the ideation.

4. Feature Origin Classification

The following classification describes the major ACIS features and the role of AI assistance. Where AI was used, outputs were reviewed and modified by the team before integration.

Feature Name	Origin	Description / AI tools, use, output and modification
1. Hybrid retrieval pipeline	Both	AI tools: ChatGPT / Claude-based coding assistance. Representative prompts asked for architecture alternatives for semantic, lexical and identifier-aware code retrieval, candidate-pool construction, and ranking. AI outputs were used as design/implementation suggestions; the team selected and modified the approach to match the Samsung Theme 1 requirements. The implemented pipeline uses

		GTE ModernBERT as the primary dense channel, BM25, exact identifiers, candidate union, ranking features, and verified code retrieval.
2. Learned ranking with LightGBM LambdaRank	Both	AI assistance was used for implementation guidance, feature/ranking design discussion, debugging, and interpretation of evaluation results. The final system uses the project implementation of LightGBM LambdaRank with 400 trees and 29 ranking signals, trained/evaluated on the project's 5,000-query development setup. The team validated the measured behavior.
3. Retrieval models and model selection	Both	AI assistance was used to compare retrieval-model options and reason about CPU/resource constraints. The final project uses Alibaba-NLP/gte-modernbert-base as the serving primary dense channel; Qwen3-Embedding-0.6B is pinned as an optional auxiliary dense channel. Encoder selection was based on the project's recorded 5,000-query CPU evaluation rather than AI recommendation alone.
4. Version-aware retrieval (P1)	Both	AI assistance supported implementation planning, debugging, and explanation of immutable snapshots, content-addressed reuse, incremental builds, atomic activation, and rollback. The team implemented and validated the actual versioning behavior, including reuse of unchanged units and safe activation of a new snapshot.
5. Evolution / lineage retrieval (Bonus)	Both	AI assistance supported architecture reasoning, implementation/debugging, and presentation of lineage concepts. The final system searches across versions, aligns revisions using evidence such as exact/moved/edited content, groups revisions into lineages, and can present one representative answer with its version history. The team validated the lineage behavior and the resulting grouped-versus-flat evaluation.
6. Evaluation and reproducibility layer	Both	AI assistance was used to interpret metrics, design checks, debug evaluation infrastructure, and draft documentation. The final evaluation records are produced by the ACIS evaluation pipeline and stored in the project's ledger/artifacts; the team reviewed the outputs and used the recorded measurements in the UI and submission material.
7. Web UI / demonstration interface	Both	AI assistance was used for UI structure, wording, component ideas, and iterative refinement. The team integrated and modified the interface to expose Search, Versions, Evolution, Evaluation, and Architecture views and to display actual ACIS results and provenance.

5. Ethical & Compliance Confirmation

AI usage complies with applicable guidelines and policies: YES

No proprietary or copyrighted data was knowingly misused through AI tools: NO

The team retained responsibility for the final architecture, implementation choices, integration, testing, evaluation, and submitted claims. AI assistance did not replace validation against the project specification or measured system outputs.

6. Declaration & Sign-Off

We declare that the information above accurately describes the use of AI assistance in developing ACIS. The final system and submitted technical claims were reviewed by the team.

Name of Team Representative	Oviya B
Role	Team Leader
Signature / Date	Oviya 04/10/26B